```cpp
#include <iostream>
using namespace std;

struct node
{
    int data;
    node *prev;
    node *next;
};

int main()
{
    node *n1 = new node();
    node *n2 = new node();
    node *n3 = new node();

    n1->data = 10;
    n2->data = 20;
    n3->data = 30;

    n1->prev = NULL;
    n2->prev = n1;
    n3->prev = n2;

    n1->next = n2;
    n2->next = n3;
    n3->next = NULL;

    node *head = n1;
    node *temp;


    temp = head;
    while (temp != NULL)
    {
        cout << temp->data << " <-> ";
        temp = temp->next;
```

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}
cout << "NULL" << endl;

// Insertion at Start
node *n4 = new node();
n4->data = 5;

n4->prev = NULL;
n4->next = head;
head->prev = n4;
head = n4;

cout << "After Insertion at Start: ";
temp = head;
while (temp != NULL)
{
    cout << temp->data << " <-> ";
    temp = temp->next;
}
cout << "NULL" << endl;

// Insertion after 20
node *n5 = new node();
n5->data = 25;

temp = head;
while (temp->data != 20)
    temp = temp->next;

n5->prev = temp;
n5->next = temp->next;
temp->next->prev = n5;
temp->next = n5;

cout << "After Insertion after 20: ";
temp = head;
while (temp != NULL)
```

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{
    cout << temp->data << " <-> ";
    temp = temp->next;
}
cout << "NULL" << endl;

// Insertion at End
node *n6 = new node();
n6->data = 40;

n6->next = NULL;

temp = head;
while (temp->next != NULL)
    temp = temp->next;

n6->prev = temp;
temp->next = n6;

cout << "After Insertion at End: ";
temp = head;
while (temp != NULL)
{
    cout << temp->data << " <-> ";
    temp = temp->next;
}
cout << "NULL" << endl;

// Deletion from Start
temp = head;
head = head->next;
head->prev = NULL;
delete temp;

cout << "After Deletion from Start: ";
temp = head;
while (temp != NULL)
```

```
{
    cout << temp->data << " <-> ";
    temp = temp->next;
}
cout << "NULL" << endl;

// Deletion after 20
temp = head;
while (temp->data != 20)
    temp = temp->next;

node *del = temp->next;

temp->next = del->next;
del->next->prev = temp;

delete del;

cout << "After Deletion after 20: ";
temp = head;
while (temp != NULL)
{
    cout << temp->data << " <-> ";
    temp = temp->next;
}
cout << "NULL" << endl;

// Deletion from End
temp = head;
while (temp->next != NULL)
    temp = temp->next;

temp->prev->next = NULL;
delete temp;

cout << "After Deletion from End: ";
temp = head;
```

```
    while (temp != NULL)
    {
        cout << temp->data << " <-> ";
        temp = temp->next;
    }
    cout << "NULL" << endl;

    return 0;
}
```